



MyTradingPal

Human-Computer Interaction 2025-2026

Phase 1 - User and Task Analysis

3LEIC03 G05

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Project's idea description

MyTradingPal is a mobile-first investment application that combines AI-driven portfolio management with interactive, character-based learning.

Users can choose to invest real money or start in a risk-free simulation mode, guided by AI characters that represent distinct trading personalities such as conservative, balanced, or aggressive. Each character provides plain-language explanations for its investment decisions, helping users understand not just what is happening, but why.

Designed for a generation of young, mobile, and curious investors, the app makes investing accessible, engaging, and transparent. It features real-time performance monitoring, explainable AI recommendations, and customizable risk settings.

Strong emphasis is placed on trust, privacy, and education through biometric authentication and privacy modes for public use.

Ultimately, the platform empowers users to learn, experiment, and grow their wealth confidently, bridging the gap between financial literacy and real-world investing through an AI experience that feels both personal and intelligent.

Related apps/services/systems

Several existing applications and platforms served as inspiration and reference points for our project. Among the most relevant are eToro, Robinhood and Trading212, each offering features that align with certain aspects of our concept while also revealing gaps our solution aims to fill.

eToro is a popular social trading platform that allows users to invest in real assets and copy the portfolios of professional traders. It demonstrates the appeal of community-driven investing and the power of accessible trading tools. However, its focus on imitation over understanding limits its educational value, providing little insight into why investment decisions are made.

Robinhood, known for commission-free trading and a simple mobile interface, set the standard for accessibility and real-time trading experiences. While it successfully opened financial markets to a broader audience, it lacks transparency and educational support, often encouraging impulsive behavior rather than informed decision-making.

Trading212 combines real-money trading with a demo mode for practice, making it an important reference for our own dual-mode design. It appeals to both beginners and experienced investors but offers limited personalization and no character-based interaction, resulting in a more transactional experience than an engaging learning one.

Our project differentiates by combining the strengths of these systems: real money investing, educational guidance, and automation into a single platform. By introducing AI characters with distinct personalities, we humanize the investing process, turning financial learning into an interactive and personalized experience. Unlike existing tools, our platform emphasizes transparency, trust, and engagement, explaining every AI decision in plain language while allowing users to experiment, learn, and invest confidently in real markets or safe simulations.

Questionnaire highlights

The user questionnaire provided valuable insights into the target audience's demographics, behaviors, expectations, and pain points, guiding both design and technical priorities of the project.

- 81% aged 18–24, 71% students: Young, learning-focused audience familiar with gamified and social apps.
- 45% beginners: Low financial literacy; require plain language and guided learning.
- 84% mobile users: Mobile-first design with one-handed, touch-friendly interaction is essential.
- 28% skeptical of automation: Explainable, transparent AI needed to build trust.
- 87% expect daily updates: Real-time notifications and dynamic feedback drive engagement.
- High privacy concern: Discreet viewing modes and biometric security required.

Overall Takeaway:

Users are young, curious, and mobile.

They want to learn safely, invest confidently, and trust AI characters that explain decisions clearly and accessibly.

PACT Analysis

People

The app targets users aged 18–45, mainly students (71%) aged 18–24, with varying investment experience — 45% beginners, 36% intermediate, and 19% experienced. Users are mobile-first and accustomed to gamified, social interfaces.

Cognitively, they face challenges understanding financial terminology and require visual, simplified explanations. 28% are skeptical of automated systems, making transparency and control vital.

Physically, the app must support one-handed, touch-based use with large, accessible elements. Socially, motivations range from learning to profit and peer comparison, but privacy remains critical.

Design Implications:

- Implement plain language throughout with contextual tooltips for unavoidable financial terminology
- Create progressive onboarding with gamified first-run tutorials that build confidence
- Establish clear visual and functional separation between simulation and real-money modes
- Provide user-adjustable UI complexity levels to accommodate varying expertise
- Ensure comprehensive accessibility: scalable fonts and high-contrast modes
- Build trust through transparent AI explanations and user control primacy
- Design for one-handed mobile operation

Activities

Users perform five main actions: monitor portfolios, customize AI characters, configure AI portfolios, run simulations, and review recommendations.

Usage is frequent: 58% check multiple times weekly, 87% expect daily updates. Sessions range from 30-second glances to 15-minute analyses.

Tasks vary in complexity, requiring glanceable dashboards, guided workflows, and confirmations for high-risk actions. Social engagement is optional but supports learning and motivation.

Design Implications:

- < 3 seconds dashboard comprehension with clear visualizations.
- Explainable AI summaries with expandable details.
- Sandbox mode for risk-free simulation.
- Explicit confirmations and decision history logs.
- Smart, non-intrusive notifications.
- Optional community features with privacy controls.

Contexts

Users access the app mostly on mobile (84%), in public or semi-private spaces, requiring discreet design. Connectivity can be unstable, so offline caching and data sync are essential.

The student demographic expects integrated tutorials, learning progression, and safe experimentation.

As a regulated real-money app, compliance with GDPR, audit trails, and financial disclaimers is mandatory.

Design Implications:

- Adaptive mobile layouts and lighting modes.
- Offline viewing with freshness indicators.
- Efficient caching, compression, and synchronization.
- Privacy mode and minimal notification previews.
- Integrated learning features and compliance workflows.
- Cross-device sync and continuity.

Technologies

The system combines real-time APIs, an AI/ML investment engine, and secure databases within a microservices architecture.

- **Frontend:** React Native apps + responsive PWA.
- **Integrations:** Broker APIs, payment gateways, biometric login (90%), and push notifications.
- **Security:** AES-256 encryption, 2FA, and session management ensure user protection.
- **Accessibility:** WCAG 2.1 AA compliance.

Design Implications:

- < 2 seconds load times, optimized charts, and lazy loading.
- End-to-end encryption and audit logging.
- Real-time WebSocket data with offline fallback.
- Accessible, scalable, and performance-monitored system.
- A/B testing and analytics for continuous improvement.

Personas

Persona 1

Name: Sofia Mendes

Quote: “I want to learn how to invest without feeling overwhelmed.”

Age: 25 years

Education level: Bachelor’s degree in Business Administration

Work/occupation: Junior Marketing Analyst

Family: Single, no children

Location: Porto, Portugal

Tech proficiency: Moderate — comfortable using apps and social media, but not experienced with financial tools

Preferred devices: iPhone and laptop

Archetype: “The learner”

Personality: dreamer, curious, motivated, cautious

Narrative / Lifestyle / Behavioural patterns: Sofia has always been curious about investing but finds the terminology and complexity intimidating. She follows finance influencers and watches YouTube tutorials but hasn’t yet taken the leap into real investing.

She appreciates tools that make learning visual, interactive, and easy to understand. In her free time, she uses apps for personal growth, budgeting, and productivity

Objectives / Needs:

- Learn how to invest safely and progressively
- Experiment with different strategies without financial risk
- Understand the impact of her decisions through clear visual feedback
- Receive AI-driven guidance and insights

Frustrations / Pain points:

- Financial terminology and complex interfaces
- Lack of personalized feedback in most trading apps
- Difficulty staying motivated when results take time
- Uncertainty about trusting automated (AI) investment tools

Persona 2

Name: Pedro Soares

Quote: “I need tools that match my pace and let me stay in control.”

Age: 34 years

Education level: Master’s degree in Computer Engineering

Work/occupation: Freelance Software Developer

Family: Married, one child

Location: Lisbon, Portugal

Tech proficiency: High experience with data, algorithms, and digital tools

Preferred devices: Laptop, tablet, and smartphone

Archetype: “The strategist”

Personality: analytical, ambitious, competitive, tech-savvy

Narrative / Lifestyle / Behavioural patterns: Pedro has been trading stocks and cryptocurrencies for several years. He enjoys testing new strategies, following financial news, and optimizing performance. With a busy work and family life, he values automation and efficiency, preferring systems that can act on his behalf while still letting him adjust decisions.

He seeks tools that combine depth and flexibility, providing clear visualizations, quick insights, and opportunities to experiment with AI-driven strategies.

Objectives / Needs:

- Automate part of his trading process using AI
- Design and customize characters that reflect his investment style
- Compare different strategies and track their performance
- Receive real-time insights and feedback on results
- Save time without losing control over key decisions

Frustrations / Pain points:

- Too many disconnected trading platforms
- Slow or unclear feedback on performance
- Difficulty visualizing risk in an intuitive way
- Lack of transparency in automated (AI) decision-making

Activity scenarios

Activity scenario 1

Sofia has just returned home after work. She decides to open her new investment application that uses AI and characters with different personalities. After logging in, Sofia chooses the “Conservative” character, curious to see how it will manage her first simulation. She sets an initial amount of 100 euros and lets the AI start investing automatically. The app displays real-time updates and small messages from the character explaining the decisions being made.

After a few weeks, Sofia checks her portfolio again and notices that it has grown by several euros. Excited with the results, she decides to sell part of her investments to lock in some profit. The app guides her through the selling process, showing how the transaction will affect her overall balance. She feels proud of her first success and starts planning to test an “Aggressive” character next time to explore a different strategy.

Activity scenario 2

After finishing a coding project, Pedro opens MyTradingPal to check on “AlphaVibe,” the custom AI character he created to trade automatically. He reviews the performance graphs and sees that AlphaVibe adjusted its portfolio after a recent market drop, keeping returns steady.

Curious to optimize further, Pedro tweaks a few parameters — lowering leverage and adding exposure to renewable energy stocks. The AI immediately simulates the new setup and shows improved long-term stability.

Satisfied with the results, Pedro lets AlphaVibe continue managing part of his portfolio. Encouraged by the app’s clear insights and automated performance, he plans to design a second, more aggressive character to compare strategies.

Functionalities

Core Functionalities

Portfolio Monitoring:

- View real-time performance of assets, gains/losses, and balance
- Quick, glanceable dashboard with visual summaries and trend indicators
- Historical charts for performance tracking and comparisons

AI Character Interaction:

- Each AI persona represents a trading style (e.g., conservative, balanced, aggressive)
- Explainable AI: characters justify recommendations in plain language
- Real-time feedback as characters adjust strategies

Personalization and Customization:

- Customize character appearance, tone, and strategy behavior
- Set personal risk levels, investment goals, and learning objectives
- Adjust notification preferences and dashboard layout

Simulation and Learning Mode:

- Sandbox mode for risk-free strategy testing
- Educational guidance through AI commentary and tutorials
- Mode separation: clearly distinguishes simulation from real-money trading

Decision Management:

- Explicit confirmation dialogs for real-money actions
- Decision history with “why it worked/failed” insights

Supportive Functionalities

Privacy and Security:

- Privacy mode to hide financial data in public settings
- Biometric login (Face ID/Fingerprint) and end-to-end encryption
- Automatic session timeout and secure logout

Offline Portfolio Access:

- View last-synced data when offline
- Automatic synchronization when connection is restored
- Data freshness indicator showing last update time

Explainable and Transparent AI:

- All recommendations include simple natural-language reasoning
- Optional “See details” for deeper insights into decisions
- Builds user trust and understanding of AI behavior

Accessibility and Usability:

- Scalable text and high-contrast themes
- One-handed mobile navigation and large touch targets

Notifications and Summaries:

- Smart daily summary notifications instead of constant alerts
- Customizable notification content and timing

Annexes

Questionnaire

<https://forms.gle/CgBJgayGpRhUpBQP6>

Results

<https://docs.google.com/document/d/1CPXZpOW-sKNJAcUpiZMqwdZBekDM-GQQMAq6Y71kKiM/edit?usp=sharing>